

Supplementary Material

Governing Societies of AI Agents: Committed Minorities,
Model Diversity, and Scale as Levers of Multi-Agent AI Governance

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This supplement holds material that supports but is not central to the argument of the main paper: a self-contained mechanical study of institutional decision rules, the full institutional-ladder arms, the granularity account of commons survival, the sustainability–equity trade-off, a methodological caution about token budgets in scale studies, and supporting figures and mappings whose values are stated in the main text. Every figure and table here is generated from the released result files at <https://github.com/kalyanar/governing-societies-of-ai-agents>.

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C.1 Institutional rules set the capture threshold

The capture threshold is not an intrinsic property of a society; rather, it is determined by the operative decision rule. In a Granovetter-style cascade framework where a committed subset of advocates promotes a change against an established status quo, free agents adopt the change once observed assembly support exceeds their individual adoption thresholds, and the system is initialized with 400 seeds the fraction of committed minority agents required to *capture* the collective decision is highly sensitive to the institutional rule (see the institutional-rules figure in Appendix A of the main paper). Under simple majority rule, capture occurs at approximately $p_c \approx 30\%$. A 2/3 supermajority rule increases resistance to capture, raising the threshold to roughly 37.5%. By contrast, *delegation* (liquid democracy, in which free agents delegate their vote to the advocate bloc) dramatically reduces the effective capture threshold to about 7.5%. A *veto* or consensus rule allows a single committed defender (approximately 1.7% of the assembly) to capture the decision by persistent obstruction.

An LLM council style confirmation setting ($N=7$, with gpt-4o-mini and Claude-Haiku monocultures versus a mixed council, initialized with 5 seeds) introduces an important qualification: sufficiently capable councils *do not become persuaded* by the reckless proposal. Free members systematically reject the unsafe option regardless of the number of advocates, implying that capture only occurs when the advocate bloc alone is large enough to mechanically satisfy the decision rule, and that a supermajority requirement prevents capture even in this case. In this regime, homogeneous and heterogeneous councils exhibit essentially identical behavior, because each individual member is sufficiently competent to converge independently on the prudent decision. This is consistent with the main text, where council composition influenced open-ended *stance diversity* but had negligible effect on decisions for which there is a clearly prudent alternative. The aggregation rule is one institutional lever; the *enforcement* mechanism is another—Bell and Madiseti [2026] show that restitution (returning a sanctioned agent’s gains) contains a defecting minority where deterrence alone fails, complementary to the voting rules studied here.

Governance implication: the choice of decision rule must be made deliberately. Supermajority requirements increase resistance to capture (at the expense of reinforcing the status quo), whereas

Table 1: Institutional layers, $N=5$, 30-month horizon, 3 seeds per cell. Every cell is clean (0/3995 decisions defaulted across all arms).

Society	0 def.	1 (20%)	2 (40%)	3 (60%)
Blind, self-interested	holds	dies 16–28	dies 6–7	—
Monitored, self-interested	holds	dies 10–12	dies 4–7	—
Monitored, <i>steward</i>	holds	holds	dies 8–12	—
+ <i>restitution</i>	holds	holds	holds	holds

delegation and veto-style rules create powerful capture channels for a very small, committed minority.

C.2 Institutional ladder: full arms

Each institutional layer buys tolerance of one more defector. Blindness and self-interest are design choices, not properties of LLM agents, and Ostrom’s field studies of durable human commons identify precisely what they omit: mutual monitoring and graduated sanctions Ostrom [1990]. We therefore add them one at a time (the table below), holding the world model fixed. *Monitoring* reveals each agent’s catch from the previous month. *Stewardship* replaces the self-interested objective with one directed at the resource persisting and the catch being shared fairly. *Restitution* returns an over-harvester’s excess to the resource before regeneration.

Two results stand out. First, *monitoring alone is mildly harmful*: making defection visible without any means of redress accelerates collapse from 16–28 months to 10–12. Information is not a governance control. Second, the objective is the first lever that works. Given the compensatory catch of the collapse-time law—6.0 tons at a stock of 80 with one defector—steward agents select exactly 6.0 where self-interested agents select their fair share of 8.0. Adding restitution extends tolerance to three defectors in five while retaining 80–92% of the adversary-free yield. A steward society therefore does possess a genuine capture threshold, bracketed by our measurements between 20% (absorbed indefinitely) and 40% (collapse within 8–12 months); with restitution the bracket moves to between 60% and 80%. At $N=5$ these brackets are one defector wide and we do not claim a point estimate inside them.

C.3 Token budget confounds scale

Token budget confounds scale. A methodological caution for anyone running such probes: reasoning-tuned models must be given room to finish. Measured at a 150–500 token cap, our 24–31B candidates scored 0.21–0.38; re-measured at 2,500 tokens with an otherwise identical prompt, the same models scored 0.71–1.00 with a 1.00 parse rate throughout. The entire apparent deficit was truncation before the answer marker. Scale studies that hold `max_tokens` fixed across a generation boundary will therefore report a capability curve that is partly an artifact of the budget.

C.4 Absolute restraint versus adversary share

The governing quantity is absolute restraint, not adversary share. The committed-minority fraction does not organize these results. A society of five survives one defector (20%) while a society of ten dies at two (20%); a 40% minority is survivable with restitution and fatal without; 60% is survivable where 80% is not. Sorting instead by the per-agent catch that the collapse-time law demands separates every observed outcome without exception (the table below): each of the seven configurations requiring more than 4.4 tons survives, and each of the

Table 2: Every steward configuration measured, ordered by the restraint the collapse-time law requires. Survival tracks the absolute catch, not the adversary share.

N	def.	share	restitution	required catch	→ outcome
5	0	0%	—	10.00	holds
5	1	20%	✓	9.50	holds
5	2	40%	✓	8.67	holds
5	1	20%	—	7.50	holds
5	3	60%	✓	7.00	holds
10	1	10%	—	4.44	holds
10	2	20%	—	3.75	dies
5	2	40%	—	3.33	dies
10	3	30%	—	2.86	dies
5	4	80%	✓	2.00	dies
10	4	40%	—	1.67	dies

five requiring less than 3.8 dies. The boundary is an absolute quantity, and it coincides with the granularity at which these agents express a decision—they answer in whole and half tons, so a target of 2.5 or 3.3 tons is one they cannot reliably hit, while 7.5 is.

This also explains the effect of scale. Because the required catch falls as $1/N$, a twenty-agent society must hold each member to 2.5 tons even with no adversary present—below the expressible floor—and it collapses within 4–6 months at $p = 0$. Group size, defector count and enforcement are not independent risks; they trade against one another on the single axis of how much restraint each agent must articulate.

Governance interpretation: the tolerable committed minority is a property of institutional design rather than of the agent population, ranging in our measurements from none at all (blind, self-interested) to three in five (steward with restitution). A share cap is therefore the wrong control here—the defence our opinion-dynamics results in the main text motivate does not transfer. What transfers is Ostrom’s pairing: an objective that makes the resource’s persistence the agent’s own aim, and an enforcement rule that reclaims what a defector takes, which is the mechanism Bell and Madisetti [2026] find necessary where informal pressure fails. Our monitoring-only arm shows why informal pressure is insufficient. A third lever is specific to artificial agents and has no direct analogue in Ostrom’s field settings: because the binding constraint is the granularity at which restraint can be expressed, widening the action space—permitting fractional allocations rather than whole units—should raise tolerance without changing the institution at all.

C.5 Sustainability and equity

Sustainability and equity trade against one another. Survival is not the only governance outcome, and the institution that secures it does so at a distributional cost. Because a steward absorbs the defector’s excess by taking less itself, the defector’s *relative* share rises: a single defector among five takes 36% of the total catch in the blind society but 44% in the steward society, a capture ratio of $1.80\times$ its population share rising to $2.20\times$, with the Gini coefficient over per-agent catch moving from 0.18 to 0.25. The honest majority is, in effect, subsidizing the adversary in order to keep the resource alive. Restitution recovers only part of this ($2.17\times$ at one defector), because a sanctioned defector is still permitted $1.2\times$ the sustainable share by construction. Two points follow. First, a minority captures more than its proportional share at *every* fraction we tested, from $1.15\times$ at four defectors in five to $2.20\times$ at one in five, so there is no fraction below which distribution is safe either—capture in the distributional sense has no threshold any more than collapse does. Second, an institution should be evaluated

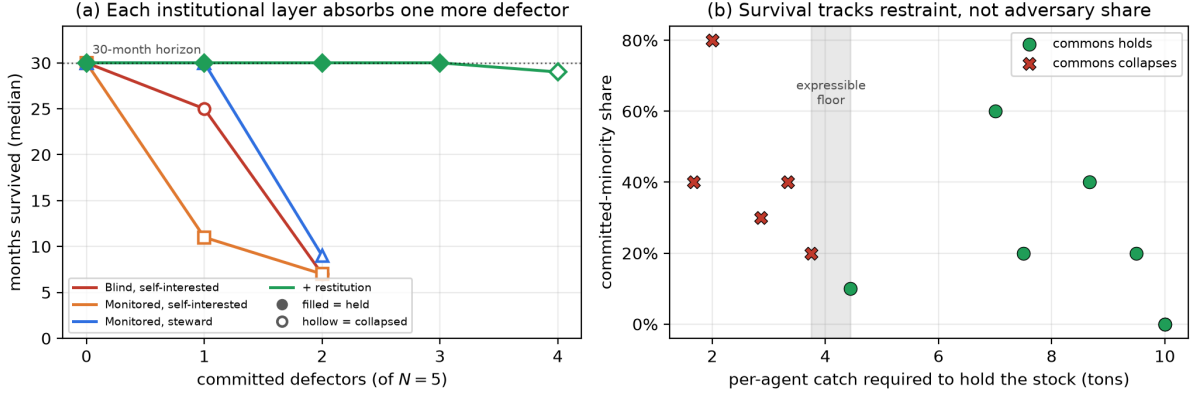


Figure 1: Commons cooperation under successive institutional layers. (a) A blind, self-interested society is destroyed by any persistent defector; monitoring alone accelerates the collapse; a stewardship objective absorbs one defector; adding restitution absorbs three in five. Survival is predicted not by the adversary’s share but by whether the restraint of the collapse-time law exceeds the granularity at which agents can express a decision.

against both objectives: stewardship alone buys sustainability while worsening equity, and it is *enforcement*, not objective-setting, that defends the honest majority’s share. We report these as indicative—three seeds per cell, with defector identity inferred from per-agent catch rather than recorded—and a dedicated distributional study is left to future work.

C.6 Value-deliberation stance spread

C.7 Injection propagation

C.8 Mapping to frameworks and standards

References

Phil Bell and Vijay K. Madiseti. Restitution over deterrence: Formal enforcement for LLM commons governance. In *Proceedings of the Second Workshop on Technical AI Governance Research (TAIGR), International Conference on Machine Learning (ICML)*, Seoul, South Korea, 2026.

Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge, UK, 1990.

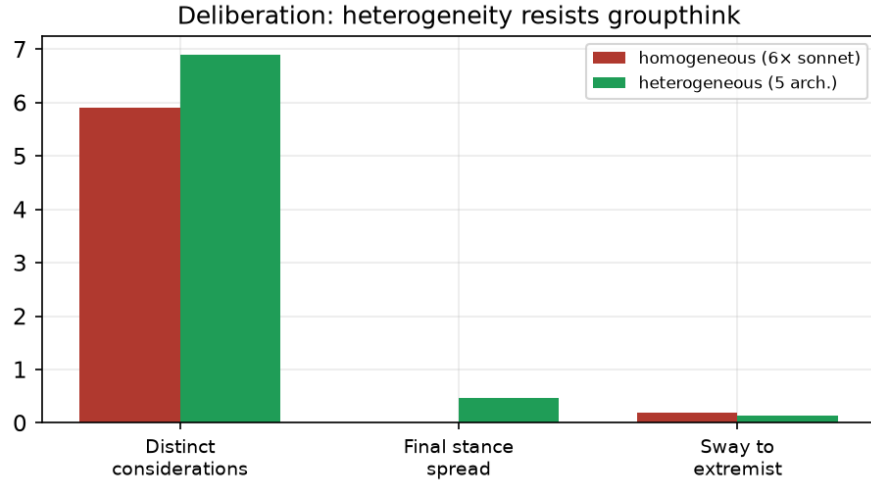


Figure 2: Value deliberation: a heterogeneous decision-making panel mitigates groupthink dynamics and resists convergence toward the position of a highly committed extremist, whereas a monocultural panel rapidly collapses to uniform attitudinal states.

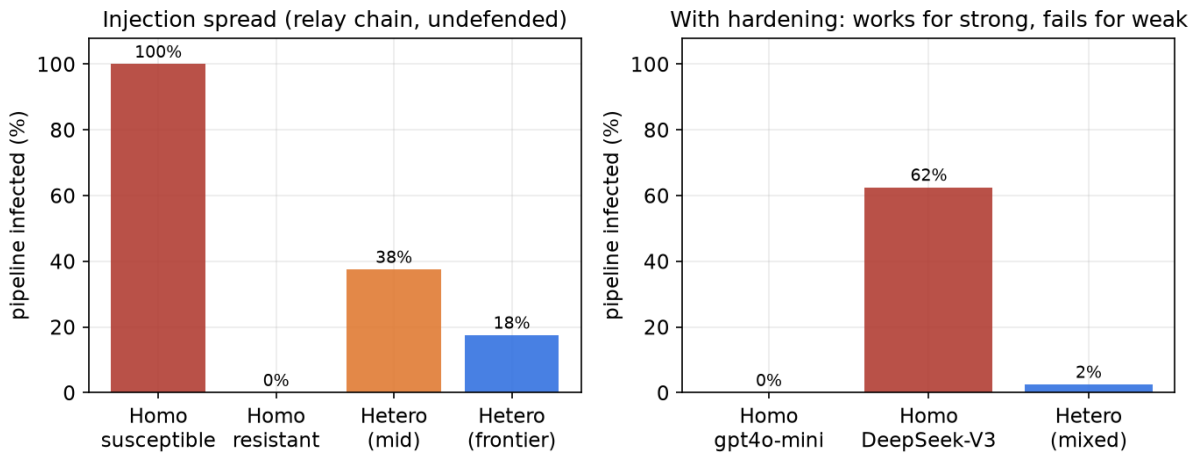


Figure 3: Prompt-injection propagation through a 6-agent relay chain. Left: a susceptible monoculture is fully compromised while a heterogeneous chain is contained by its first immune link. Right: hardening works for strong models but fails for robustly-susceptible ones; diversity plus hardening is best.

Table 3: Mapping findings to AI-governance frameworks and standards.

Finding	Framework	Provision
Injection propagation; mono-culture attack surface (see the main text)	OWASP; EU AI Act	LLM01 Prompt Injection; Art. 15 (robustness & cybersecurity)
Capture threshold as quantifiable attack surface (see the main text)	NIST AI RMF; EU AI Act	MEASURE/MANAGE (secure & resilient); Art. 9 (risk mgmt)
Vendor diversity; non-uniform patching (see the main text)	NIST GenAI Profile; EU AI Act	Value-chain & component integration; provider due-diligence
Capability gate / fitness to govern (see the main text)	EU AI Act	Art. 15 (accuracy); Art. 9 (risk mgmt)
Capture monitoring; pre-cascade plateau (see the main text)	EU AI Act; NIST AI RMF	Art. 14 (human oversight); MANAGE
Auditing the aggregation rule (see the main text)	NIST AI RMF	GOVERN (accountability)
Verifier as engineered control (see the main text)	NIST AI RMF	MEASURE (TEVV)